UTDesign EPICS - Fall 2026

ENGINEERING DECISION RECORD (ADR)

EPCS 2200 / EPCS 3200 | Team Deliverable | 100 Points

Purpose

An ADR documents why the team made an important engineering decision. It captures the problem, options considered, evidence used, decision made, and the tradeoffs or consequences that follow.

When to create an ADR

Create an ADR for a meaningful engineering decision - not for routine task choices. Examples include architecture, material or component selection, sensor choice, platform/framework choice, protocol/interface, manufacturing approach, algorithm, build-vs-buy, major redesign, or another decision with significant technical consequences.

Grading and continued use

One significant ADR is graded for 100 points at the course milestone. After that, teams should create additional ADRs whenever meaningful engineering decisions occur. Individual ADRs remain stable records; if a later decision replaces an earlier one, create a new ADR and mark the prior ADR as superseded.

1. Decision Identification

What to submit: Provide the basic information needed to identify and track this decision.

Project Name	BrainCharge – Software Team
Decision Title	Library choices for noise cancellation feature
Date	10/1/2026
Author(s) / Contributors	Sreeja Amaresam
Decision Owner	Sharat
Project Type	New / Carry-over
Status	Proposed / Accepted / Superseded / Deprecated

2. Context / Decision Problem

What to submit: Explain the engineering decision that needs to be made and why it matters to the project. Include enough context for someone who was not in the discussion to understand the issue.

Keep this focused. Do not retell the entire project history.

The compassion robot currently requires a conversational aspect, and part of this feature means that the robot needs to be able to decipher user speech so that the LLM can process it to give a response back. The problem is that in real-life, the audio that the robot will hear will have background noise from conversations and other activities. We need a piece of software that will help filter out this background noise so that the robot can be more reliable to the user. Thus far, the Jetson that is responsible for the computing aspects gets an input through its microphone. The software that is responsible for transcribing the user speech currently does a good job at capturing the words accurately, although this fails during presentation times where there are other conversations going on the background. We are simply adding a new program that filters the audio.

New vs. Carry-over

New project: the ADR may document a foundational engineering choice needed to move the design forward. Carry-over project: clearly state what was inherited from prior teams and whether the current team is retaining, modifying, replacing, or revalidating an earlier design decision.

3. Decision Drivers / Constraints

What to submit: List the most important factors the team must consider when making this decision. These should explain what makes one option more suitable than another.

Examples: performance, cost, safety, compatibility, maintainability, usability, schedule, fabrication limits, available tools, sponsor requirements, integration needs, or relevant requirement IDs.

Decision Driver / Constraint	Why It Matters
RAM	We only have a limited capacity for RAM, and this limits the capability of our programs and how many programs can run at the same time.
Compatibility across OS platforms	The Jetson Nano runs on Linux and the environments available for testing with all members (our laptops) are either Windows or macOS. The program should be compatible with all environments so that we can write code, test, and run the program.
Latency	The robot should respond quickly to have an effective conversation.
Price	The solution should ideally be cheap or no-cost because we have already reached a high cost for the robot.

4. Options Considered

What to submit: Identify the reasonable alternatives the team considered. Include the advantages and disadvantages or tradeoffs of each option.

Do not force a certain number of alternatives. Consider the reasonable options appropriate to the decision. If only one viable option remains, explain why the others were eliminated.

Option	Advantages	Disadvantages / Tradeoffs
WebRTC Audio Processing Module (APM)	It applies echo cancellation (AEC), noise suppression (NS), and VAD all at the same time. It has been heavily tested with VoIP applications, and is compatible with Linux, macOS, Windows. Although it is written in Rust/C++, we can use something like Meson to integrate it into our python codebase.	The VAD has low accuracy (especially in noisy conditions), which is what we do not want. There are also no benchmarks or prototype results we can compare this too (for the entire module at least; there are benchmarks done for the VAD component).
Hardware mic with onboard DSP	It applies echo cancellation (AEC), noise suppression (NS), and VAD all at the same time on the hardware (so RAM or GPU is not impacted).	Requires purchasing additional hardware, which may be impractical given our sponsor's goals to keep the price as low as possible.

Option	Advantages	Disadvantages / Tradeoffs
SileroVAD + RNNoise from xiph/rnnoise	These two libraries together provide AEC and noise reduction. SileroVAD provides high accuracy and has regular updates from the developers. RNNoise provides real-time noise suppression with low latency.	RNNoise must be compiled separately for each OS type and is built for a unix-style shell.

5. Evidence / Analysis

What to submit: Summarize the evidence used to compare the options. Use only evidence relevant to this decision. Attach or link supporting calculations, tests, prototypes, specifications, simulations, benchmarks, or other records when useful.

Software - Examples if Applicable

Prototype results • benchmarks • API/platform compatibility • performance • security • maintainability • deployment constraints • integration testing

Hardware - Examples if Applicable

Calculations • simulation • measurements • vendor specifications • material properties • load/functional testing • fabrication constraints • safety • component availability

Mixed projects may use evidence from both. Complete only what is relevant.

WebRTC:

Compatibility - https://gitlab.freedesktop.org/pulseaudio/webrtc-audio-processing/-/blob/master/meson.build?ref_type=heads

This is the link to the meson.build file on the official repository on Gitlab. This is evidence for its compatibility via Meson.

https://en.wikipedia.org/wiki/WebRTC

The Wikipedia entry for WebRTC shows that it supports the following browsers:

- Desktop PC
    - Microsoft Edge 12+
    - Google Chrome 28+
    - Mozilla Firefox 22+
    - Safari 11+
    - Opera 18+
    - Vivaldi 1.9+
    - Brave
- Android
    - Google Chrome 28+ (enabled by default since 29)
    - Mozilla Firefox 24+
    - Opera Mobile 12+
- ChromeOS
- Firefox OS
- BlackBerry 10
- iOS
    - MobileSafari/WebKit (iOS 11+)

- Tizen 3.0

Maintainability - https://gitlab.freedesktop.org/pulseaudio/webrtc-audio-processing/-/commits/master?ref_type=HEADS (last commit made on November 10, 2025)

Hardware Mic with DSP:

Performance and platform compatibility: https://wiki.seeedstudio.com/respeaker_xvf3800_introduction/

The link shown above is the documentation for the Seeed ReSpeaker XVF3800, which is a potential microphone that we can use that has built-in features such as “AEC, AGC, DoA, beamforming, VAD, noise suppression, de-reverberation”. It’s specifications include:

- **Quad Microphone Array** :4 high-performance mics in circular pattern for 360° far-field voice capture up to 5 meters
- **Advanced Audio Processing** :Powered by XVF3800 with AEC, multi-beamforming, de-reverberation, DoA detection, dynamic noise suppression, 60dB AGC range
- **Open Source Compatible** :Works with USB hosts (Windows, macOS, Raspberry Pi OS) and I2S hosts

Deployment constraints: Since we have less than half of our budget remaining for this project, spending more on a specialized microphone may not be feasible based on cost. Currently, this specific microphone sits between $50 and $80 in price.

https://www.amazon.com/seeed-studio-reSpeaker-XVF3800-Linear-4/dp/B0H94ZZN99. This is one of the products we could purchase, and it is $73.

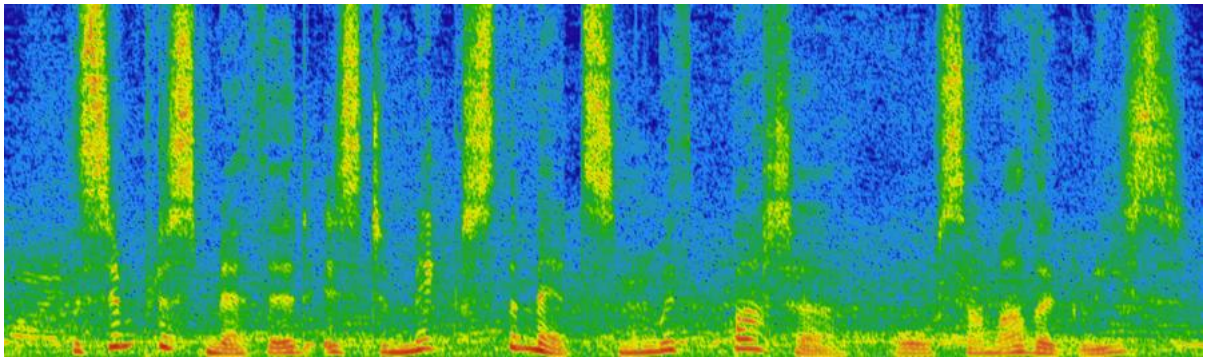
Third party information that verifies the information above can be found here: https://smarthomecircle.com/respeaker-xvf3800-home-assistant-voice-assistant

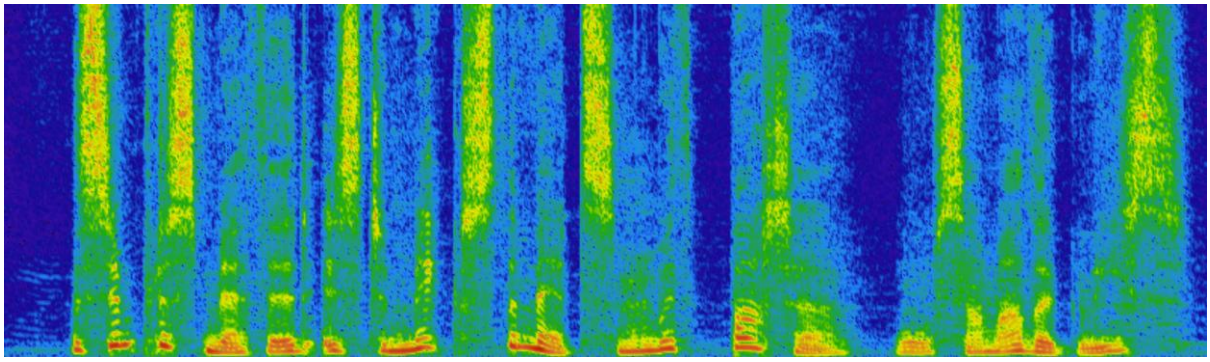
The site validates the features provided by the microphone, such as the processor capabilities with noise suppression, the microphone array, etc.

It also shows that it has a good hearing detection range at 3 meters.

Silero VAD + RNNoise:

Benchmarks:





The image above shows the spectrogram of a piece of audio before and after running the noise suppression from RNNoise. There is a visible difference in the amount of visual noise with cleaner lines/audio segments.

API/platform compatibility:

Silero-vad is available as a python library at https://pypi.org/project/silero-vad/

RNNoise:

https://gitlab.xiph.org/xiph/rnnoise/-/blob/0.1.x/configure.ac

The link above goes to the configuration file of RNNoise, which shows how different OS's need different compiled binaries. The code evidence is shown below.

Platform-specific tweaks case $host in *-mingw*) # -std=c89 causes some warnings under mingw. CC_CHECK_CFLAGS_APPEND([-U__STRICT_ANSI__]) # We need WINNT>=0x501 (WindowsXP) for getaddrinfo/freeaddrinfo. # It's okay to define this even when HTTP support is disabled, as it only # affects header declarations, not linking (unless we actually use some # XP-only functions). AC_DEFINE_UNQUOTED(_WIN32_WINNT,0x501, [We need at least WindowsXP for getaddrinfo/freeaddrinfo]) host_mingw=true ;; esac AM_CONDITIONAL(OP_WIN32, test "$host_mingw" = "true")

This can mean potential issues with having a consistent build across different OS's for each team member.

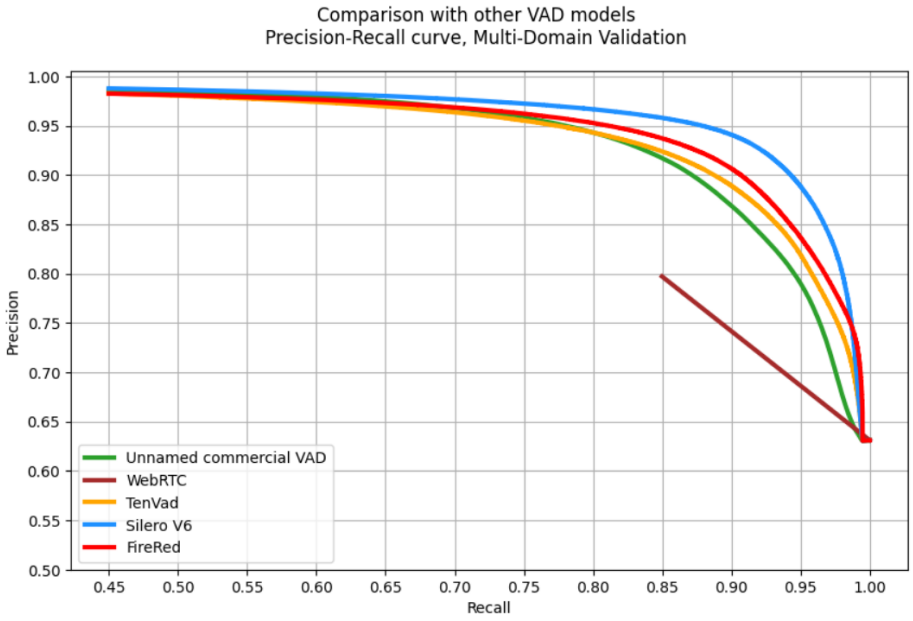
RNNoise is not compatible with Windows currently.

https://github.com/xiph/rnnoise/pull/227

https://github.com/xiph/rnnoise/pull/88

These are pull requests on the official RNNoise repository to make the program compatible to windows by removing/replacing components not compatible with MSVC. It has yet to be pushed into the repository.

Performance:



This graph is the Precision-Recall curve that compares SileroVAD with other VAD models (including WebRTC). It mainly shows the level of precision of the speech captured versus how much of the speech is captured. SileroVAD shows the strongest precision overall for different recall levels. For example, at 95% of the real speech recalled, SileroVAD had the highest precision at around 90%. Additionally, it shows WebRTC as having the worst precision (with only around 70% precision at 95% recalled). This information is from the documentation page for SileroVAD (https://github.com/snakers4/silero-vad).

Maintainability:
SileroVAD - https://github.com/snakers4/silero-vad (had a commit update on September 29, 2026)

6. Decision

What to submit: State clearly what the team decided.

We have decided to use SileroVAD and RNNoise to get the best audio filtration possible given the project's limitations. If working with RNNoise is not feasible, we will use only SileroVAD.

7. Rationale

What to submit: Explain why the selected option was chosen over the reasonable alternatives. Connect the explanation to the decision drivers, constraints, and evidence above.

SileroVAD + RNNoise is the better option compared to WebRTC and a separate mic that filters audio because of a few factors. We have a big constraint on cost. The robot was requested to be made as accessible as possible for potential users (which would be caregivers), and adding another specialized product would increase the cost for the consumer. We won't go with WebRTC either because it does not have the best accuracy and capabilities according to the precision-recall curve.

8. Consequences / Tradeoffs

What to submit: Describe what this decision improves, limits, costs, changes, or requires. Include important tradeoffs the team is accepting.

A strong engineering decision rarely has only benefits. Make the important tradeoffs visible.

We will probably need to sacrifice even more of our existing current functionality to meddle with the VAD system (we currently already have one with Whisper.cpp). If we decide to go with RNNoise as well, we would need to spend more time to check if the program is compatible with our devices (though WSL systems or such), which would take some time away from developing features. It would improve the accuracy of the speech that is recorded by the robot in most reasonable noisy conditions (such as background conversations).

9. Follow-Up / Validation

What to submit: Identify any actions, tests, procurement, integration work, sponsor discussion, or later validation needed because of this decision.

Follow-Up Action	Owner	Target / Checkpoint
Get both software on a Windows laptop and a linux system to assess compatibility and what the build would look like.	Sharat	Make sure that the program works and runs without issue, especially on Linux.
Run the conversation software and check for acceptable latency and accuracy.	Sharat	Ensure the user will have a reasonable response time (within 3 seconds).

Follow-Up Action	Owner	Target / Checkpoint
Run the program with the robot itself (and the Jetson Nano)	Sharat	Ensure it works in practice on the actual robot with good latency (response within 3 seconds) and accuracy (above 90%).

10. Mentor Review

What to submit: Review the decision with the mentor as part of normal project guidance. The mentor may challenge alternatives, evidence, assumptions, tradeoffs, or whether additional validation is needed.

Date Reviewed	October 8, 2026
Review Status	Reviewed / Needs Additional Analysis
Important Mentor Guidance	Don't focus too much on Windows if it becomes a problem. WSL2 is an option and it may be viable to disable/bypass the noise cancelation in windows environments

Submission Checklist

Before submitting the graded ADR, confirm that:

- ☐ The ADR documents a meaningful engineering decision, not a routine task choice.
- ☐ The decision problem and why it matters are clear.
- ☐ Carry-over projects clearly identify any inherited design decision relevant to the current choice.
- ☐ The important decision drivers or constraints are identified.
- ☐ Reasonable alternatives were considered rather than jumping directly to a preferred option.
- ☐ The comparison uses relevant engineering evidence or analysis.
- ☐ The decision and rationale are explicit.
- ☐ Important consequences and tradeoffs are visible.
- ☐ Follow-up or validation actions are identified where needed.
- ☐ The ADR has been reviewed with the mentor.

How the EPICS Artifacts Fit Together

Artifact	Primary Question	What It Contains
Project Brief	What are we solving?	Problem, stakeholders, requirements, constraints, semester target, success criteria
Project Plan	How will we execute?	Roles, workstreams, milestones, dependencies, risks, near-term priorities
ADR	Why did we make this important engineering choice?	Options, evidence, decision, rationale, tradeoffs, follow-up

Core design principle

Requirements tell us what the solution must achieve. The Project Plan tells us how the team will execute the work. ADRs explain why important engineering choices were made.

ADR Grading Rubric - 100 Points

Team grade: The graded ADR is one shared team deliverable. The same artifact, rubric, and team score apply to EPCS 2200 and EPCS 3200. Course-level differences are assessed through individual contribution/performance expectations.

Criterion	Points	What Strong Work Demonstrates
Context & Decision Framing	15	The engineering decision is meaningful, clearly framed, and grounded in relevant project context. Carry-over projects identify inherited decisions where applicable.
Decision Drivers & Constraints	15	The factors that matter to the decision are relevant, clear, and connected to project needs, constraints, or requirements.
Alternatives Considered	20	Reasonable alternatives are identified and compared with meaningful advantages, disadvantages, and tradeoffs.
Evidence & Analysis	25	The team uses appropriate engineering evidence, analysis, testing, calculations, prototypes, specifications, or other relevant information to support the comparison.
Decision, Rationale & Tradeoffs	20	The selected option is explicit, the reasoning is defensible, and important consequences or tradeoffs are acknowledged.

Criterion	Points	What Strong Work Demonstrates
Clarity & Professional Quality	5	The ADR is concise, understandable, traceable, and useful as a future engineering record.

TOTAL: 100 points